GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

Custom Motor Interface and Drawing Checklist

For OEM designers specifying a motor when standard sizes do not fit the equipment

Work through the sections in order. Section 1 defines the problem, Sections 2 to 5 build the drawing package, and Section 6 sets the checks for sample approval. Use the effort table to choose the lowest rung that solves each item.

Step 1 — Define what does not fit

- [ ] List each failing item with its number, e.g. body 12 mm too long, shaft engagement 6 mm short
- [ ] Classify each item as geometric, performance or integration
- [ ] Record the standard motor you compared against and its data sheet revision
- [ ] State which machine dimensions are fixed and which can still change
- [ ] Record annual volume and whether the part is for prototypes, series or spares

Step 2 — Shaft interface

- [ ] Diameter with tolerance class and extension length from the mounting face with tolerance
- [ ] Key width × height × length, or flat depth and length, thread, cross-hole or centre hole
- [ ] Single or double-ended shaft; rear extension length if an encoder, brake or handwheel is fitted
- [ ] Mating part: coupling, pulley, pinion or sprocket, and its distance from the shaft shoulder
- [ ] Peak torque the drive can deliver (stall torque or current limit × Kt) for the shaft check
- [ ] Radial and axial load at the shaft and its position, for the bearing check

Step 3 — Mounting and envelope

- [ ] Mounting type: B3 foot, B5 flange or B14 face, plus required orientation
- [ ] Spigot diameter and depth with fit, PCD, hole count, hole size or thread
- [ ] Maximum length, diameter or square section, including connector and cable bend
- [ ] Keep-out zones, service access to brushes or connectors, and airflow paths
- [ ] STEP model of the mating parts and envelope, plus a 2D drawing with datums

Step 4 — Electrical and connection

- [ ] Supply voltage: nominal, minimum and maximum at the motor terminals
- [ ] Load torque and speed at the operating point, peak torque and duty type (S1, S2, S3)
- [ ] Lead exit as a clock angle relative to a mounting hole, cable length and bend radius
- [ ] Connector part number, pin-out and required IP rating at the connector
- [ ] Ambient temperature at the motor and any thermal protector requirement

Step 5 — Integrated options and sealing

- [ ] Brake: fail-safe or spring-applied, coil voltage, holding torque, manual release
- [ ] Encoder or Hall sensors: resolution, output type, supply voltage
- [ ] Gearhead: ratio, output torque including peaks, output shaft and mounting

- [ ] IP code required; note that seals add friction and reduce cooling
- [ ] EMI suppression and any electronic control to be integrated

Step 6 — Tolerance stack-up and sample approval

- [ ] List every contributor to each critical fit with its nominal and ± tolerance
- [ ] Worst case: minimum = nominal – sum of tolerances; use for safety-relevant fits and small batches
- [ ] RSS: minimum = nominal – square root of the sum of squared tolerances; use for stable high-volume processes
- [ ] Fit-check samples on the real machine; measure shaft runout and spigot engagement
- [ ] Retest back-EMF constant, resistance, no-load current and temperature rise after any change
- [ ] Release against a frozen drawing revision with written acceptance limits

Customisation effort levels and key checks

Customisation	Effort	Key check
Shaft length, flat, key, thread	Low	Torsion at stall torque; overhang bearing load
Shaft diameter	Low–medium	Stress rises with $1/d^3$; bearing bore limit
Double-ended shaft	Low–medium	Added length; rear seal
Lead exit, cable, connector	Low	IP rating and strain relief at exit
Flange, spigot, PCD	Medium	Concentricity; adaptor length
Winding for voltage or speed	Low–medium	Retest Ke, resistance, temperature rise
Stack length	Medium	Torque falls roughly with stack length
Brake, encoder, gearhead	Medium	Length; brake voltage; gearbox rating
Sealing upgrade	Medium	Seal friction; temperature rise
New lamination or housing	High	Volume or hard envelope limit justifies tooling

Effort levels describe engineering and tooling change, not price or lead time, and depend on the specific frame. Confirm every customisation against a dimensioned drawing and a sample test before series release.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.